THE NORMALIZATION FACTORIZATION OF A UNIVERSALLY CLOSED QUASI-SEPARATED MAP OF SPACES, FLAT-BASE-CHANGE COMPATIBLE

THE STACKS PROJECT AUTHORS

Editorial note. Statement span is lines 7993-8017 of sources/stacks-spaces-more-morphisms/pinned-original-source.tex; proof span is lines 8019-8058, the complete author proof (38 body lines).

Editorial note. Navigation stubs: 7 cited-but-not-included labels are supplied as non-load-bearing pointers with their pinned tags where the snapshot tag index has them.

Editorial note. extra_packages is empty on purpose: the retained preamble already loads xy with options and a second declaration clashes in the assembler wrapper.

Editorial note. Text is retained verbatim, including the author's own typos and placeholders.

Editorial note. Counted B side (both complete). Pillar consumed by both Stein theorems L16200/L16203 and the lsc card L16212 — all one wave, chain closes.

Allowed standard prerequisites (editorial).

- Explicit prerequisite boundary. Morphisms of Spaces: universally-closed-quasi-compact, normalization-in-universally-closed, universally-closed-permanence, quasi-compact-permanence, compose-after-separated, pushforward; Morphisms of Spaces definition normalization-of-Y-in-X.

Source reference. Original lemma 0822: sources/stacks-spaces-more-morphisms/pinned-original-source.tex, source lines 8058–8058. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

Source reference. Original lemma 03KR: sources/stacks-spaces-more-morphisms/pinned-original-source.tex, source lines 8058–8058. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

Source reference. Original lemma 0A0Q: sources/stacks-spaces-more-morphisms/pinned-original-source.tex, source lines 8058–8058. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

Source reference. Original lemma 03M9: sources/stacks-spaces-more-morphisms/pinned-original-source.tex, source lines 8058–8058. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

Source reference. Original lemma 03KS: sources/stacks-spaces-more-morphisms/pinned-original-source.tex, source lines 8058–8058. Cited by the retained statement or

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proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

Source reference. Original lemma 04NX: sources/stacks-spaces-more-morphisms/pinned-original-source.tex, source lines 8058–8058. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

Source reference. Original lemma 04XW: sources/stacks-spaces-more-morphisms/pinned-original-source.tex, source lines 8058–8058. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

Lemma 0.1. *Let S be a scheme. Let $f : X \to Y$ be a universally closed and quasi-separated morphism of algebraic spaces over S. There exists a factorization*

$$\begin{array}{ccc} X & \xrightarrow{\quad} & Y' \\ & {\scriptstyle f'} \searrow \quad \swarrow {\scriptstyle \pi} & \\ & {\scriptstyle f} \searrow \quad \swarrow & \\ & Y & \end{array}$$

with the following properties:

1. *the morphism f' is universally closed, quasi-compact, quasi-separated, and surjective,*
2. *the morphism $\pi : Y' \to Y$ is integral,*
3. *we have $f'_*\mathcal{O}_X = \mathcal{O}_{Y'}$,*
4. *we have $Y' = \underline{\mathrm{Spec}}_Y(f_*\mathcal{O}_X)$, and*
5. *Y' is the normalization of Y in X as defined in Morphisms of Spaces, Definition 0822.*

Formation of the factorization $f = \pi \circ f'$ commutes with flat base change.

Proof. By Morphisms of Spaces, Lemma 04XW the morphism f is quasi-compact. We just define Y' as the normalization of Y in X, so (5) and (2) hold automatically. By Morphisms of Spaces, Lemma 0A0Q we see that (4) holds. The morphism f' is universally closed by Morphisms of Spaces, Lemma 04NX. It is quasi-compact by Morphisms of Spaces, Lemma 03KS and quasi-separated by Morphisms of Spaces, Lemma 03KR.

To show the remaining statements we may assume the base Y is affine (as taking normalization commutes with étale localization). Say $Y = \mathrm{Spec}(R)$. Then $Y' = \mathrm{Spec}(A)$ with $A = \Gamma(X, \mathcal{O}_X)$ an integral R-algebra. Thus it is clear that $f'_*\mathcal{O}_X$ is $\mathcal{O}_{Y'}$ (because $f'_*\mathcal{O}_X$ is quasi-coherent, by Morphisms of Spaces, Lemma 03M9, and hence equal to $\widetilde{A}$). This proves (3).

Let us show that f' is surjective. As f' is universally closed (see above) the image of f' is a closed subset $V(I) \subset Y' = \mathrm{Spec}(A)$. Pick $h \in I$. Then $h|_X = f^\sharp(h)$ is a global section of the structure sheaf of X which vanishes at every point. As X is quasi-compact this means that $h|_X$ is a nilpotent section, i.e., $h^n|X = 0$ for some $n > 0$. But $A = \Gamma(X, \mathcal{O}_X)$, hence $h^n = 0$. In other words I is contained in the Jacobson radical of A and we conclude that $V(I) = Y'$ as desired. □